

BUFFALO HARBOR, NY, USA

WMO: 997254

Lat: 42.878N

Lon: 78.890W

Elev: 585

StdP: 14.39

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	2.5	7.1	N/A	N/A	N/A	N/A	N/A	N/A	40.2	28.7	35.6	27.8	13.2	270	0.557

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.7	80.2	N/A	78.4	N/A	77.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	7.9	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	(o)	(p)
(4) 32.6	27.4	24.1	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			DB	-1.4	84.7	7.0	2.3	-6.5	86.4	-10.5	87.7	-14.5	89.0	-19.5	90.7	
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	48.4	25.3	24.6	31.9	43.7	54.8	65.5	71.1	70.2	63.7	53.2	42.0	32.7
	DBStd	18.09	10.67	9.75	8.93	7.27	7.25	5.41	4.11	4.02	5.91	7.23	8.25	8.24
	HDD50	3137	766	712	565	214	33	0	0	0	0	49	261	537
	HDD65	6576	1230	1132	1027	640	326	58	5	6	93	370	689	1000
	CDD50	2537	0	0	3	25	182	465	653	627	411	149	21	1
	CDD65	502	0	0	0	0	10	72	192	168	54	6	0	0
	CDH74	1428	0	0	0	3	19	155	609	490	146	6	0	0
	CDH80	71	0	0	0	0	1	17	25	21	7	0	0	0
Wind	WSAvg	10.4	13.6	12.1	10.2	9.1	7.7	8.1	8.7	8.9	9.1	11.4	12.6	13.6
Precipitation	PrecAvg	40.5	2.9	3.0	3.1	3.1	3.1	3.7	2.7	3.7	4.0	3.3	4.3	3.9
	PrecMax	49.1	5.6	6.1	6.0	5.0	6.9	6.3	5.5	6.6	7.7	5.6	9.6	6.3
	PrecMin	32.3	1.0	0.4	0.8	1.6	0.4	1.1	0.9	0.5	1.4	0.8	1.4	1.9
	PrecStd	4.4	1.4	1.7	1.4	1.0	1.5	1.4	1.4	1.6	1.7	1.5	2.0	1.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.7	49.2	61.2	69.5	77.0	82.0	83.2	82.4	80.5	74.1	63.3	55.5
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	47.6	43.8	54.2	63.3	71.9	77.9	80.1	79.7	77.5	70.2	59.7	50.4
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	43.7	40.3	48.3	59.2	68.8	75.4	78.4	78.1	75.5	67.4	56.9	47.4
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	39.8	36.7	43.9	55.3	66.0	73.1	77.2	76.8	73.5	64.3	54.4	44.1
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	11.1	11.2	11.4	13.2	11.9	9.7	8.7	9.6	11.7	10.9	10.3	9.2
		MCDBR	14.3	14.2	17.4	21.0	16.6	14.6	9.8	11.0	12.8	14.8	13.2	12.2
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub	0.298	0.328	0.334	0.375	0.404	0.437	0.437	0.427	0.380	0.362	0.315	0.299	
	taud	2.395	2.301	2.396	2.325	2.285	2.227	2.240	2.258	2.370	2.409	2.507	2.459	
	Ebn at Noon	266	274	288	283	276	266	265	264	269	259	260	256	
	Edh at Noon	25	32	33	38	41	44	43	41	34	29	22	22	
All-Sky Solar Radiation	RadAvg	413	665	1035	1395	1711	1843	1859	1652	1315	795	493	331	
	RadStd	39	66	114	149	179	141	144	91	121	76	70	44	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (52 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page